Machine Learning Based Early Detection of Cattle Diseases

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Abstract*—Mastitis, Foot-and-Mouth Disease (FMD), Lumpy Skin Disease (LSD), and Milk Fever (hypocalcaemia) continue to hit Sri Lanka's dairy farmers hard, mostly because they are not caught in time: veterinary officers are not always within reach in rural areas, so by the time a sick animal is examined, milk yield has already dropped. We present CattleSense, a multimodal machine learning platform that lets farmers screen for all four conditions through a single web application, with no lab access or veterinary login required. Each module is built around the data that helps diagnose its disease. Mastitis pairs a ResNet50 image classifier with a Decision Tree on milk biomarkers; FMD uses a fine-tuned EfficientNetB0 lesion classifier alongside a weather-derived outbreak-risk signal; LSD chains a YOLOv8s nodule detector with a ResNet50 classifier and an optional symptom checklist; and Milk Fever risk comes from a Random Forest–XGBoost ensemble adjusted for heat stress via a live Temperature-Humidity Index. On held-out test data, the system reached 87.0–99.4% accuracy across modules (AUC 0.94–0.999), giving farmers clear risk levels and practical next steps to shorten diagnostic delays.

Keywords-Mastitis, Foot-and-Mouth Disease, Lumpy Skin Disease, Milk Fever, Cattle Disease Detection

I. Introduction

Cattle farming underpins much of Sri Lanka's rural economy, with roughly 1.2 million cattle supporting dairy production and household income for smallholder farmers nationwide. That livelihood keeps taking hits from the same handful of diseases - Mastitis, Foot-and-Mouth Disease, Lumpy Skin Disease, and Milk Fever - each of which, caught late, means lost milk yield, rising treatment costs, and in the case of FMD and LSD, spread across the herd.

Diagnosis still comes down mostly to manual observation checking udders, skin, or watching a freshly calved cow for weakness. It's subjective and often too slow: vets and labs aren't evenly spread across the country, so farmers in outlying areas can wait days for an assessment that early-stage disease doesn't allow. FMD and LSD make this worse, since both spread fast through contact or insect vectors - one missed animal can turn into a herd-wide outbreak.

Machine learning offers a way to screen earlier and more reliably - CNNs catch visual cues like lesions and inflammation, while ensemble models handle structured indicators like biomarkers and activity level that farmers can observe without lab access. Neither covers everything alone, which is why this work treats each disease as its own hybrid, multimodal problem.

This paper presents CattleSense, a four-module platform built collaboratively - Mastitis, FMD, LSD, and Milk Fever - sharing a common web interface and cattle-record database, with a joint discussion following each module's results.

II. Literature Review

Automated livestock diagnostics has drawn steady research attention, with deep learning and multimodal fusion of visual, clinical, and environmental signals at the centre of recent work. Manual veterinary inspection is repeatedly identified as the bottleneck in dairy herd management, especially where veterinary access is scarce and diagnostic delays are common. We review prior work under each of the four diseases.

A. Mastitis Detection

Machine learning and deep learning have increasingly been applied to bovine mastitis detection using both image-based and sensor-derived features [1], [2]. CNN architectures applied to udder thermal images have achieved accuracies above 87% [3], while ultrasonography-based deep networks have been explored for distinguishing mastitis-affected tissue in buffaloes [4]. Sensor-driven approaches using milk conductivity, somatic cell count, and biomarker measurements have also shown reliable classification through ensemble models [5]. Most studies rely on a single modality, motivating multimodal fusion approaches combining visual and physiological signals for more robust detection [6].

B. Foot-and-Mouth Disease (FMD) Detection

Automated visual diagnosis of FMD - targeting lesions on the mouth, tongue, and hooves - has drawn considerable deep learning attention. A YOLOv4-based detector reported accuracy as high as 98% on annotated lesion images [7], while an ensemble of VGG16, ResNet50, and InceptionV3 achieved 98.2% across multi-country datasets [8]. Work here also extends beyond image classification - interpretable ML models have been used to study FMD epidemiology [9], flag

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incidence on farms lacking veterinary access [10], and connect seasonal and environmental factors to outbreak risk across South Asia [11].

C. Lumpy Skin Disease (LSD) Detection

Deep learning has reached livestock skin conditions in general but not LSD specifically. CNNs classified mixed cattle skin diseases at over 85% accuracy [12], and VGG16/InceptionV3 transfer learning has been applied to skin lesion detection [13]; neither targeted one disease or a farmer-facing tool. Foundational epidemiology documented LSD's presentation and transboundary spread across Africa, the Middle East, and Asia [14], [15], yet diagnosis has stayed manual. YOLOv8 [16] and ResNet50 [17] are each well proven, but no published pipeline combines them for LSD.

D. Milk Fever Detection

Milk Fever is common metabolic issue affects dairy cows around calving. It occurs when the cow's blood calcium level decreases during the calving period. Recent studies have investigated how to predict hypocalcemia in dairy cows using Matching Learning [19] used farm and animal related data with machine learning models to predict periparturient hypocalcemia. [21], [18] found that changes in lying behavior and activity were associated with hypocalcemia in dairy cows during the transition period. [20], [22] Their study evaluated XGBoost, LSTM models , behavioral information can support early hypocalcemia prediction. Most existing approaches depend on sensors, lab measurements, or detailed farm records. This module takes a more accessible path, building prediction around indicators a farmer can observe directly.

III. Methodology

Each module was developed, evaluated independently, reflecting the different evidence available for each condition image data drives some, while others lean more on clinical, behavioural, or environmental indicators. All four still follow the same design pattern: a primary model trained on the strongest available signal, combined through a hybrid fusion layer with a secondary farmer-collectible data source, and evaluated on a held-out test set never seen during training.

A. Mastitis Detection

The Mastitis Detection module takes a hybrid approach, pairing an image-based deep learning model with a numerical clinical model. Two publicly available datasets were used for this - one sourced from Kaggle, the other from Roboflow: an udder image set (1,172 images spanning mastitis and normal classes) and a clinical biomarker dataset (800 records). Before training, the image dataset was screened for exact and near-duplicate entries using file hashing and perceptual hashing, and then split into training, validation, and test sets using a cluster-aware strategy to guard against data leakage. Every image was resized to 224×224 pixels and harmonized through fixed-quality JPEG re-encoding, with augmentation applied only on the training side.

For the image branch, a ResNet50 CNN with a frozen, pretrained backbone was trained to classify udder images as mastitis-positive or normal. Alongside this, a Decision Tree classifier was trained on five biomarkers that a farmer could plausibly collect - milk temperature, pH, conductivity, yield, and clotting - to generate an independent numerical prediction. These two outputs are then merged through a soft-voting hybrid fusion layer, which can optionally be refined using a farmer-reported symptom checklist, to arrive at the final diagnosis. Fig. 1 shows the overall architecture.

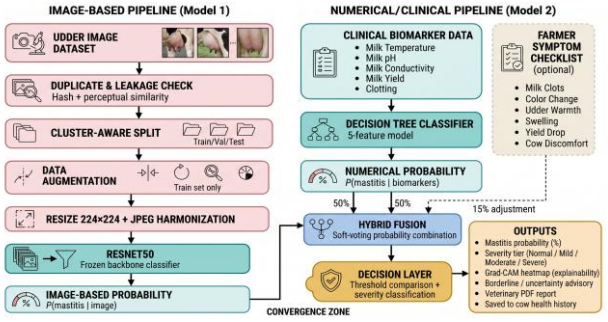


Fig. 1. Model architecture of Mastitis Detection

B. Foot-and-Mouth Disease (FMD) Detection

At the core of the FMD module is a fine-tuned EfficientNetB0 CNN trained on 339 labelled photographs of cattle mouths, tongues, hooves, and udders - 170 FMD-negative and 169 confirmed positive - manually checked to remove corrupted files. Since the raw data had no case identifier, a filename-based heuristic was used to group images taken from the same animal at multiple angles. That grouping directly shaped how the data was split: an 85/15 grouped stratified train-test split and grouped 3-fold cross-validation were used so that images from the same case could never end up on both sides of a split and leak information into the results. All images went through the same preprocessing pipeline - decoded, resized to 160×160, normalized - whether for training or live inference, keeping both paths consistent.

Fine-tuning used class-weighted loss and EfficientNetB0's own input rescaling. The final model was then evaluated on a held-out test set it had never seen, reporting accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix. That image prediction feeds into a rule-based hybrid layer alongside optional clinical symptom scores and a weather-driven spread-risk signal built from live Open-Meteo data, with a seasonal escalation rule reflecting DAPH's reported higher FMD risk between December and February. Together, these produce the final risk level and recommendations shown to the farmer. Fig. 2 lays out the full pipeline.

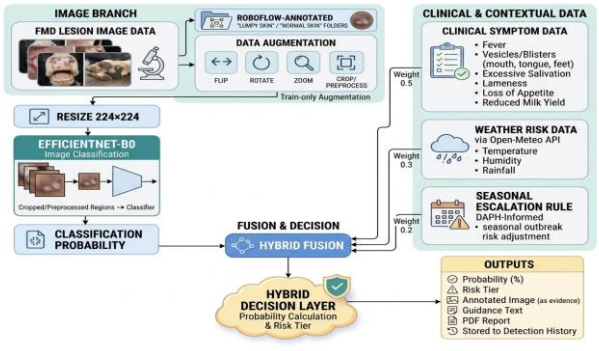


Fig. 2. Model architecture of Foot-and-Mouth Disease (FMD)

C. Lumpy Skin Disease (LSD) Detection

The LSD module combines YOLOv8s and ResNet50 in a hybrid deep learning pipeline. Two datasets were used - 254 annotated images for nodule detection and 382 images split into "Lumpy Skin" and "Normal Skin" folders for classification - both resized, normalised, and augmented on

the training set only. YOLOv8s localises skin nodules, ResNet50 classifies each detected region, and their confidence scores are multiplied per region, with the highest score across the image forming the overall photo-based probability. This feeds into a weighted hybrid fusion layer alongside an optional farmer-reported symptom checklist to produce the final risk decision. The full pipeline is shown in Fig. 3.

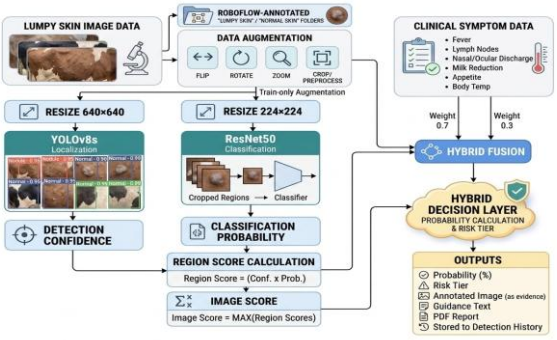


Fig. 3. Model architecture of LSD

D. *Milk Fever Detection*

The proposed milk fever detection module uses matching learning to get prediction approach. Full Combined Dataset of 8783 records gathering from three sources: A clinically grounded dataset (2000 records) made using validated veterinary rules, Areal dataset (1316 records) real world collections from Sri Lankan farm, and Real IOT Sensor Dataset (5467 records from 580 dairy cows) collected from existing GitHub repo. All records were fully standardized to eight features: parity, blood calcium, blood phosphorus, body condition score (BCS), days to calving, milk yield on day one, activity level, and dietary cation-anion difference (DCAD). SMOTE oversampling was applied to the training dataset to address class imbalance in the Critical stage, producing 2,700 samples per class. Seven machine learning algorithms trained and compared Logistic Regression, SVM, MLP Neural Network, Random Forest, XGBoost, Gradient Boosting, and an Ensemble Voting Classifier combining Random Forest and XGBoost through soft voting. The selected model was Ensemble model for production model based on its AUC-ROC of 0.9986 and validation accuracy of 98.15%. It was enabling prediction without laboratory access. The Architecture of the system is shown in Fig. 4.

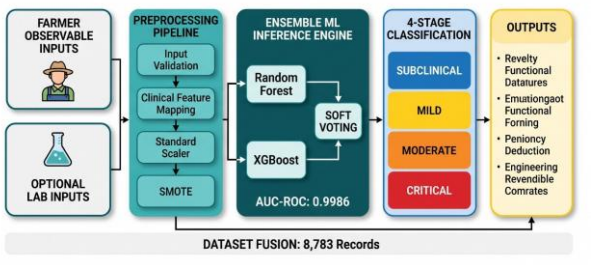


Fig. 4. Model architecture of Milk Fever Detection

IV. Results and Discussion

A. *Mastitis Detection*

We evaluated the proposed system across three components: the ResNet50 image classifier, the Decision Tree numerical model, and the combined hybrid fusion approach. The training and validation performance of each model is shown in TABLE I.

TABLE I. ACCURACY OF EXPERIMENTED MODELS

Model	Train Accuracy	Validation Accuracy
ResNet50 (Image)	93.37%	81.77%
Decision Tree (Numerical)	-	99.69%

The ResNet50 backbone was frozen and pretrained, with early stopping based on validation AUC. As Fig. 5 shows, training accuracy climbed steadily, reaching 93.37% at the best epoch, while validation accuracy settled around 81.77%. The gap points to mild overfitting expected given the limited dataset though early stopping kept it from widening further.

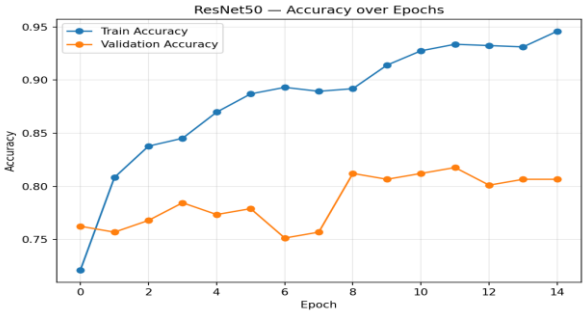


Fig. 5. Accuracy graph of ResNet50 classification model

On the held-out test set - data the model never saw during training - the final model reached 87.01% accuracy, with precision at 91.45% and recall at 89.17%. The ROC curve in Fig. 6 backs this up, showing strong discriminative ability with an AUC of 0.9376.

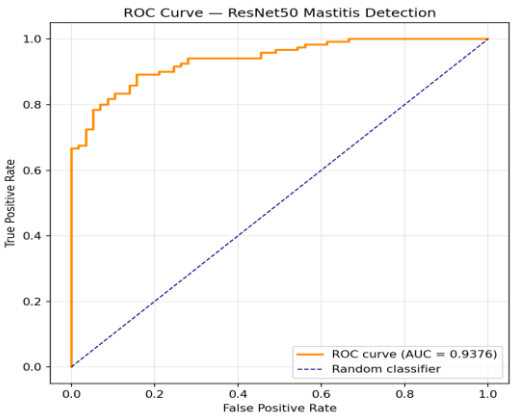


Fig. 6. ROC curve of the ResNet50 mastitis detection model

Looking at the confusion matrix in Fig. 7, the model correctly flagged 107 true mastitis cases and 47 true normal cases, with 10 false positives and 13 false negatives. This precision-recall balance was intentional - in a clinical setting, missing an actual mastitis case is far more costly than raising a false alarm, so the model was tuned with that trade-off in mind.

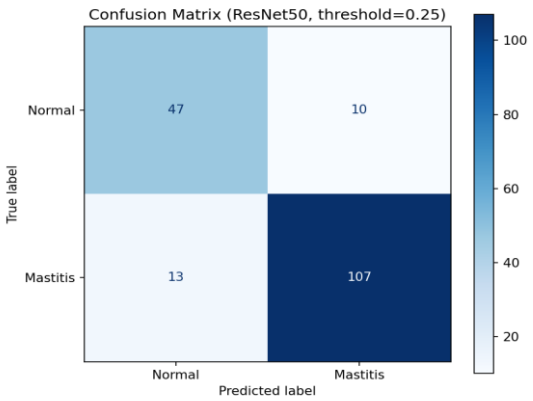


Fig. 7. Confusion matrix of mastitis detection (ResNet50)

The Decision Tree, working from five farmer-collectible biomarkers, hit 99.38% test accuracy with only one wrong call out of 160 samples. Repeated stratified cross-validation (5-fold × 10 repeats) held at 99.69% mean accuracy with a standard deviation of just 0.38% not surprising given how closely milk conductivity and temperature are known to track mastitis status.

Both models feed into a soft-voting fusion layer, giving equal weight to image-based and biomarker predictions when both are available, with an optional symptom checklist as a refinement step. When biomarker data isn't on hand, the system falls back to image-only mode. Grad-CAM heatmaps over the udder region also show which areas drove each prediction, giving farmers and vets a visual basis to trust or question a result rather than treating it as a black box.

B. Foot-and-Mouth Disease (FMD) Detection

The FMD image classifier was evaluated on a held-out test set of 51 images (25 no-FMD and 26 FMD-positive) using a grouped split to prevent data leakage. The fine-tuned EfficientNetB0 achieved 94.1% accuracy, with 94.2% precision, 94.1% recall, and 94.1% F1-score. For FMD-positive cases, it achieved 92.6% precision, 96.2% recall, and 94.3% F1-score, with an ROC-AUC of 0.971. Three-fold grouped cross-validation produced accuracies of 86.7%, 89.5%, and 95.1%, indicating stable performance.

TABLE II. MODEL PERFORMNACE SUMMURY

Model	Key Metric	Value
EfficientNetB0	Test Accuracy	94.1%
EfficientNetB0	Macro F1-Score	94.1%
EfficientNetB0	FMD Recall	96.2%
EfficientNetB0	ROC-AUC	0.971
EfficientNetB0	Avg. Precision (PR-AUC)	0.968

The training and validation accuracy of the EfficientNetB0 model across 17 training epochs is illustrated in Fig. 8. Validation accuracy converges quickly to around 92–94% within the first two epochs and remains stable thereafter, while training accuracy continues to improve gradually, indicating that the model generalizes well without severe overfitting.

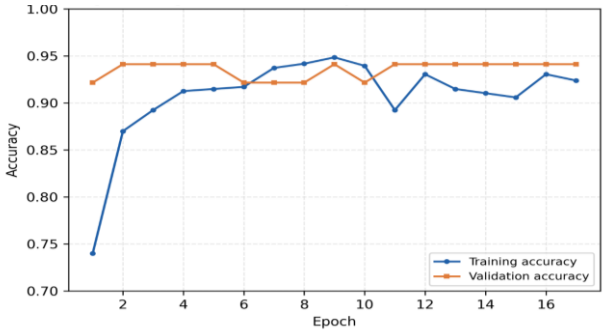


Fig. 8. Training and validation accuracy curve of the EfficientNetB0 model

The precision-recall curve, computed from the model's predicted FMD probabilities on the held-out test set, stays close to the top-right corner of the plot, confirming that the model sustains high precision even as recall increases - the desired behavior for a decision-support tool where both missed disease cases and false alarms carry real cost.

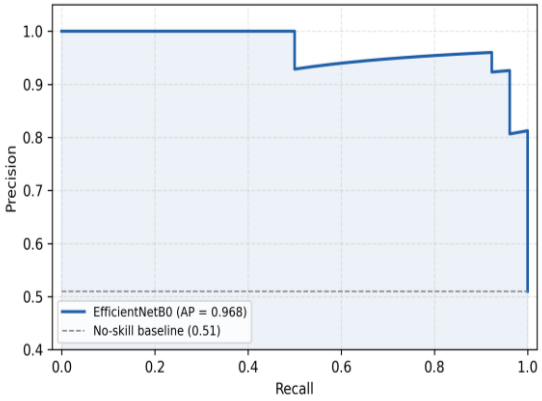


Fig. 9. Precision–Recall curve of the EfficientNetB0 FMD classifier

The confusion matrix (Fig. 10) breaks this down further: of the 25 true no-FMD images, 23 were classified correctly and 2 came back as false positives; of the 26 true FMD images, 25 were caught correctly, with just 1 false negative slipping through.

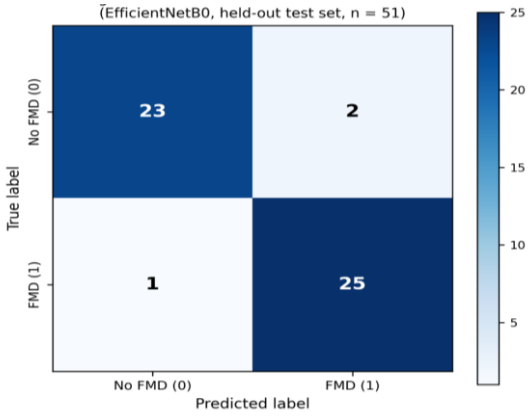


Fig. 10. Confusion matrix of the EfficientNetB0 FMD classifier

The module combines three signals into one assessment: the image model's FMD probability, an optional clinical score from temperature and symptoms, and a district-level weather-based risk score from live Open-Meteo data.

C. *Lumpy Skin Disease (LSD) Detection*

The YOLOv8s model was used to detect candidate skin nodules in cattle photographs, while ResNet50 was used to classify each detected region as LSD-positive or healthy. TABLE III summarizes the performance achieved model.

TABLE III. RESNET50 PERFORMNACE SUMMARY

Model	Key Metric	Value
ResNet50	Test Accuracy	93.10%
ResNet50	ROC-AUC	0.9637

Fig. 11 shows YOLOv8s box loss dropping steadily across 79 epochs, with precision and recall both trending upward. Best mAP@0.5 hit 13.83% at epoch 48, dropping to 11.83% on the held-out test set alongside precision of 0.19 and recall of 0.28. The modest figures reflect a genuinely harder problem than classification - nodules vary considerably in size, shape, and visibility against the coat.

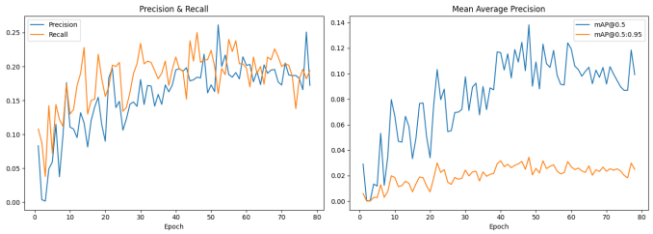


Fig. 11. YOLOv8s training curves across 79 epochs.

ResNet50, on the other hand, was trained in two phases, with accuracy and AUC curves plotted in Fig. 12. Training accuracy eventually hits 100%, while validation accuracy settles in around 95% and on the actual held-out test set, the model reaches 93.1% accuracy with an AUC of 0.9637.

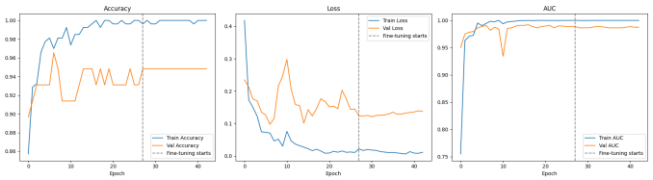


Fig. 12. ResNet50 accuracy, loss, and AUC across both training phases.

The model correctly classified most of the test subset, identifying 6 of 9 healthy cases and 48 of 49 LSD-positive cases. The confusion matrix in Fig. 13 confirms strong class separation for the LSD-positive class on this dataset.

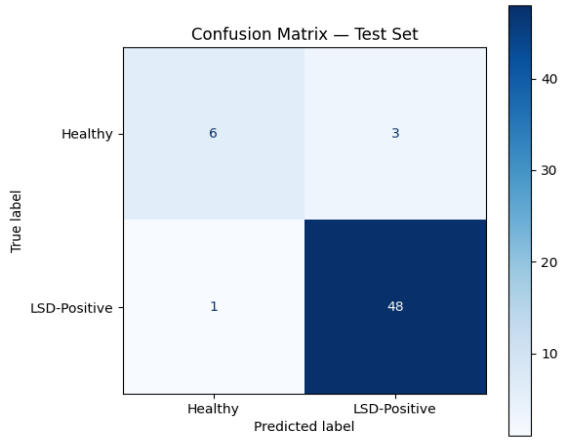


Fig. 13. Confusion matrix - ResNet50 LSD classification (test set).

Together, these results suggest that pairing nodule detection with image classification gives a workable foundation for automated LSD screening, even if the detection stage still has room to grow. An optional symptom checklist rounds things out - image-based results carry 70% of the final decision, symptom input the remaining 30% - pulling in clinical observations alongside the visual evidence.

D. *Milk Fever Detection*

The system uses an Ensemble Voting Classifier combining Random Forest and XGBoost through soft probability averaging, trained on 8,783 records from three sources - a clinically simulated set (2,000 records), a real-world Sri Lankan farm set (1,316 records), and an IoT sensor dataset (5,467 records from 580 dairy cows). Eight features served as input: parity, blood calcium, blood phosphorus, body condition score, days to calving, first-day milk yield, activity level, and dietary cation-anion difference. SMOTE oversampling was applied to the training set only to address underrepresentation in the Critical stage, balancing all four classes at 2,700 samples each.Random Forest builds 200 trees through bagging, reducing variance and handling noisy or correlated features well. XGBoost builds 200 trees sequentially at a learning rate of 0.05, correcting earlier errors and capturing complex feature relationships. Final predictions combine both models through soft voting, averaging their probability outputs so neither dominates - which matters most near the Critical stage boundary, where a wrong call carries clinical consequences. Model performance is summarized in TABLE IV.

TABLE IV. ACCURACY OF EXPERIMENTED MODELS

Model	CV Accuracy	Test Accuracy
Random Forest	96.38% ± 0.34%	95.28%
XGBoost	97.95% ± 0.31%	96.98%
Ensemble (RF+XGB)	98.15% ± 0.23%	97.04%

In the test set of 1,757 cases, the performance was 97.04% for accuracy, 97.04% for weighted F1-score, and 0.9986 for AUC-ROC, all surpassing the proposed minimum requirements of 90% and 0.90. Cross-validation held at 98.15%, confirming consistent generalization (Fig. 14).

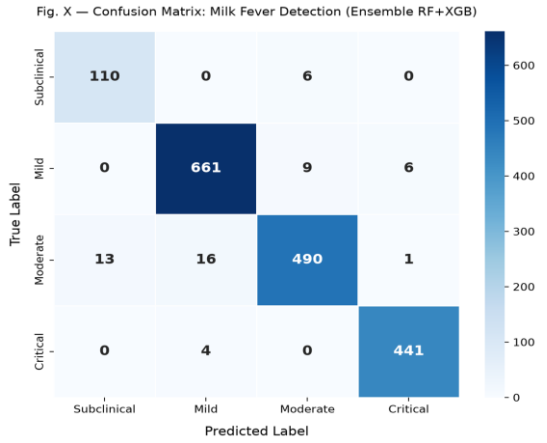


Fig. 14. Confusion matrix of Milk Fever Detection

The confusion matrix clearly shows excellent class separations. The Subclinical and Mild phases got F1-scores of 98% and 97% – this is impressive considering that the stages have no observable signs. The Critical phase got

93.97% recall and 89% precision, with errors classified in the Moderate phase instead of the lower levels, thus maintaining clinical safety. The ROC curves are shown on Fig.15.

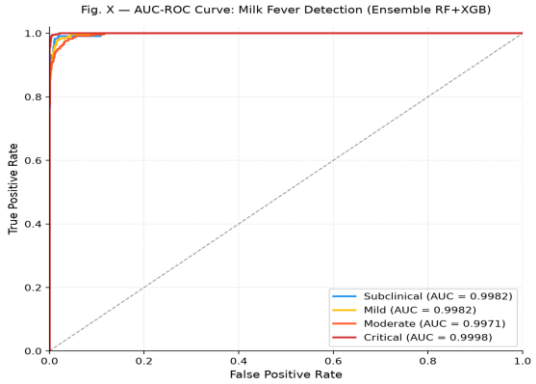


Fig. 15. AUC-ROC curve of the Milk Fever Detection

E. Overall System Performance & Validation

Across modules, accuracy on held-out test data ran from 87.0% for mastitis imaging, whose biomarker model reached 99.4%, 94.1% for FMD and 93.1% for LSD, to 97.0% for Milk Fever staging. Performance held up even where photograph quality was poor or farmer-reported data was thinner than ideal, which supports the hybrid design across all four conditions.

V. Conclution

This study set out to build CattleSense, a platform to catch four diseases hitting Sri Lankan dairy farms hardest - Mastitis, FMD, LSD, and Milk Fever - before they get out of hand. Each disease got its own hybrid pipeline pairing image-based deep learning with whatever clinical, biomarker, or environmental data made sense, rather than forcing everything through one model. Results held up well: Mastitis imaging hit 87.0% (AUC 0.94) with biomarkers reaching 99.4%, FMD 94.1% (AUC 0.97), LSD 93.1% (AUC 0.96), and Milk Fever staging 97.0% (AUC 0.999). Pairing a photo with farmer-collectible data consistently outperformed either source alone - without needing lab access or a vet login.

That's the gap all four literature reviews pointed to: no screening tool built for Sri Lankan farmers in a form they can use. CattleSense tries to close it by turning a photo or quick checklist into a risk level within seconds, helping farmers reach a vet sooner and, for the two contagious diseases, catch outbreaks before they spread. Work remains - LSD nodule detection needs more training images, and longitudinal per-animal risk tracking is still ahead. Field trials with real farms are the next step, since held-out test performance only tells part of the story.

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